Week 11 Assignment

Generalized linear models

Render this file to PDF and submit both the `.qmd` and the rendered `.pdf` via Canvas before Week 12's lecture (Monday). See the syllabus for the late policy.

```
library(tidyverse)
library(marginaleffects)
```

Part A: Binary response. Using the built-in `mtcars` data, model whether a car has a manual transmission (`am`) as a function of weight (`wt`).

1. Fit a logistic regression.
2. Use `avg_slopes()` to report the average change in the probability of a manual transmission per 1,000 lb (1 unit of `wt`). Interpret it in one sentence.
3. Plot predicted probability against weight with `plot_predictions()`.

Part B: Count response. Using `InsectSprays`:

4. Fit a Poisson regression of `count ~ spray`. Check for overdispersion.
5. Use `avg_comparisons()` to report rate ratios of each spray relative to spray A. Which sprays differ most from A?

Part C: Chi-square as a linear model. A survey of 100 plants: 40 of 60 plants in sun had flowers; 10 of 40 plants in shade had flowers.

6. Build the 2×2 table, then run `chisq.test(correct = FALSE)`.
7. Run the same test as a Poisson `glm()` with an interaction, and show the test statistics match.
8. In 2–3 sentences: explain why the interaction (rather than a main effect) is the test of association.